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CENG 407 - Conclusion & Future Work

**VR Anatomy: VR Based Educational Interactive Human Anatomy
Training Platform**

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1. Introduction

Anatomy education requires students to understand the human body not only as a collection of separate structures, but also as a three-dimensional system in which organs, bones, muscles, vessels, and other anatomical parts are connected to each other. However, traditional learning materials such as textbooks, two-dimensional diagrams, and classroom presentations may not always be sufficient for students to visualize these spatial relationships clearly. This can make anatomy difficult to understand, especially for students who are learning the subject for the first time.

Virtual Reality (VR) offers a strong opportunity to support anatomy education by allowing students to experience anatomical structures in an immersive and interactive environment. Instead of only reading descriptions or looking at static images, students can observe 3D models from different angles, interact with anatomical parts, and explore the body in a more active way. This type of learning environment can make abstract or complex anatomical concepts more concrete and easier to remember.

At the same time, artificial intelligence can support the learning process by providing explanations, answering student questions, and helping learners review topics when they need additional guidance. When combined with VR, AI can make the learning experience more flexible and supportive by allowing students to receive information during exploration rather than depending only on fixed lesson content.

The main purpose of this project is to develop VR Anatomy, a VR-based educational platform that combines interactive anatomy visualization with AI-supported learning. The project aims to create a learning environment where students can explore selected anatomical systems, receive guided explanations, ask questions, and evaluate their understanding through quizzes. In this way, the system is designed to support traditional anatomy education with a more visual, interactive, and student-centered approach.

2. Project Overview

VR Anatomy is a functional prototype developed as a graduation project to support anatomy learning in a virtual reality environment. The system was designed for students who need visual and interactive support while learning introductory anatomy topics. The primary target group of the project is high school students taking anatomy and physiology courses, and the application was tested in a real classroom context with this target group during the first experimental condition.

The current scope of the application focuses on selected topics from the musculoskeletal system and the circulatory system. These systems were selected because they include structures that benefit from three-dimensional visualization, such as bones, muscles, the heart, and vessels. Due to time, content, and implementation constraints, the project does not cover the entire human anatomy. Instead, it presents a focused prototype that demonstrates how VR and AI can be used together in anatomy education.

The system includes four main modules. The AI Guided Learning module provides a structured learning flow in which students follow guided anatomical explanations. The Free Explore module allows students to examine anatomical models independently. The AI Chat module enables

students to ask anatomy-related questions through text or voice and receive AI-supported answers. The Quiz module allows students to check their understanding after the learning process.

The project also includes avatar-based guidance and voice interaction. In the learning environment, the avatar supports the student by presenting explanations and creating a more natural interaction experience. The system includes speech-to-text and text-to-speech features so that students can communicate with the AI assistant by speaking or typing. These features aim to make the platform more interactive and accessible for different learning preferences.

From a technical perspective, the VR application was developed in Unity using XR interaction components. The artificial intelligence features were implemented through a Python-based backend that provides retrieval-supported answers from anatomy-related educational content. The backend also supports speech processing features used in the AI Chat and learning modules.

The project was also designed with two experimental interaction conditions. Experiment 1 uses a neutral learning experience with no student name collection, formal language, and a formal avatar. This condition has been completed and tested with high school students. Experiment 2 is planned as a follow-up condition that will use name-based personalization, friendlier language, and a more casual avatar. Therefore, the current version of the report presents the completed system and the initial evaluation status based on Experiment 1.

3. System Design and Architecture

The VR Anatomy system was designed as a combination of a Unity-based virtual reality application and a Python-based artificial intelligence backend. The Unity application is responsible for the visual learning environment, 3D anatomical models, user interaction, avatar presentation, quiz flow, and scene management. The backend is responsible for processing AI-related requests, retrieving relevant anatomy content, generating answers, and supporting speech-based interaction features.

The general architecture follows a client-server structure. The VR application acts as the client and sends requests to the backend when the user asks a question, requests a simplified explanation, or uses voice-based interaction. The backend processes these requests and returns the generated response to the Unity application. The response can then be displayed as text, read aloud through text-to-speech, or presented through the avatar depending on the active module.

The system was divided into multiple scenes and modules in Unity to keep the project organized and easier to manage. Each module has a specific purpose in the learning process. The AI Guided Learning module provides guided instruction, the Free Explore module supports independent model inspection, the AI Chat module handles open-ended question answering, and the Quiz module supports knowledge assessment. This modular structure made it easier to develop, test, and improve each part of the application separately.

The application uses 3D anatomical models to represent selected structures from the musculoskeletal and circulatory systems. In AI Guided Learning, the relevant structure can be highlighted while unrelated structures can be hidden or made transparent. This design helps students focus on the current topic without being visually overwhelmed by all anatomical parts at once. In free exploration, students can interact with models more independently and examine them from different angles.

For artificial intelligence support, the backend uses a retrieval-based approach. Instead of generating answers without context, the system first searches anatomy-related educational content and retrieves relevant information. Then, the AI response is generated based on this retrieved content. This design was preferred to make the answers more consistent with the learning material and to reduce the risk of irrelevant or unsupported explanations.

The system also includes speech-to-text and text-to-speech features. Speech-to-text allows students to ask questions by speaking, while text-to-speech allows the system to read answers aloud. These features were added to make the application more natural in a VR environment, where typing can be less comfortable than speaking. Voice interaction also supports the avatar-based learning experience by making the communication feel more conversational.

Overall, the architecture of VR Anatomy connects interactive VR learning with AI-supported educational assistance. The Unity side provides the immersive and visual learning experience, while the backend provides intelligent explanation and voice support. This separation allows the system to remain modular, maintainable, and expandable for future improvements.

3.1 Unity VR Application

The main application was developed in Unity because Unity provides strong support for virtual reality development, 3D scene management, physics-based interaction, and cross-platform deployment. The VR environment was designed to allow students to observe anatomical structures in an immersive space and interact with them through VR controllers.

The Unity side of the project includes the main menu, module selection screens, anatomical learning scenes, free exploration scenes, quiz screens, avatar presentation, and user interface panels. The application was organized into separate scenes so that each learning module could be managed independently. This structure helped reduce complexity during development and made it easier to test individual parts of the system.

The project uses XR interaction components to support VR controller input, ray-based selection, object interaction, and model inspection. In the learning scenes, students can use controller-based interaction to select buttons, navigate between panels, and interact with anatomical models. In free exploration scenes, students can examine selected anatomical structures more independently.

A major design goal of the Unity application was to reduce visual complexity during anatomy learning. For this reason, the system can highlight the currently explained anatomical structure while making unrelated structures transparent or less visually dominant. This helps students focus on one topic at a time and understand the location of each structure within the larger anatomical system.

The Unity application also includes avatar-based presentation. The avatar is used to make the learning experience more natural and guided. Depending on the active experimental condition, the avatar can represent either a more formal learning style or a more casual and friendly learning style. This design supports the project's experimental comparison between neutral and personalized interaction styles.

Overall, the Unity VR application forms the visual and interactive core of the project. It provides the environment where students experience anatomy through 3D models, guided explanations, free inspection, quizzes, and avatar-supported interaction.

3.2 AI and RAG Backend

The artificial intelligence component of the project was implemented as a Python-based backend. This backend is responsible for receiving requests from the Unity application, processing anatomy-related questions, retrieving relevant educational content, and returning generated answers to the VR interface.

The backend uses a retrieval-augmented generation approach. In this approach, the system does not rely only on the language model's general knowledge. Instead, it first searches the prepared anatomy learning content and retrieves relevant pieces of information. The generated answer is then formed based on this retrieved content. This makes the AI responses more connected to the course material and more suitable for an educational application.

The anatomy content used by the backend was prepared from educational sources related to the selected learning scope. The content was divided into smaller text chunks so that the system could search and retrieve the most relevant parts when a student asks a question. This structure allows the AI assistant to provide answers that are more focused on the user's question.

The backend is used in more than one part of the application. In the AI Chat module, students can ask open-ended anatomy questions and receive AI-supported answers. In the AI Guided Learning flow, the backend can also support simplified explanations when students need easier wording or additional clarification. This allows the system to provide adaptive support instead of presenting only fixed explanations.

One of the main reasons for using a separate backend is flexibility. The AI system can be improved, updated, or deployed independently from the Unity application. This also makes it possible to change the AI model, update the anatomy content, improve prompts, or add new features without rebuilding the entire VR application.

Overall, the AI and RAG backend provides the intelligent learning support layer of VR Anatomy. It helps the application move beyond static educational content by allowing students to ask questions and receive context-aware explanations during the learning process.

3.3 Speech-to-Text and Text-to-Speech Integration

Speech interaction was added to make the VR learning experience more natural and easier to use. Since typing in a VR environment can be difficult and slow, the system supports both speech-to-text and text-to-speech features. These features allow students to communicate with the AI assistant by speaking and to listen to the generated answers.

The speech-to-text feature allows students to record their questions through the microphone. The recorded voice input is sent to the backend, converted into text, and then processed as a question for the AI assistant. This makes the AI Chat module more suitable for VR use because students do not have to rely only on keyboard input.

The text-to-speech feature converts generated answers into spoken audio. After the AI assistant produces an answer, the system can display the response as text and also read it aloud. This is especially useful in a VR learning environment because students can continue looking at the 3D model while listening to the explanation.

Voice interaction is also connected to the avatar-based presentation. When the avatar presents or reads explanations, the learning experience becomes more conversational and guided. This supports the goal of making the application feel less like a static information screen and more like an interactive educational assistant.

The integration of speech-to-text and text-to-speech also supports accessibility and different learning preferences. Some students may prefer reading the text, while others may understand better by listening. By supporting both formats, the system provides a more flexible learning experience.

Overall, speech integration improves the usability of VR Anatomy by reducing the need for manual text input and by making explanations easier to follow inside the VR environment. It also strengthens the connection between the AI assistant, the avatar, and the student.

4. Implemented Modules

This section describes the user-facing functions implemented in each module of the VR Anatomy application.

4.1 AI Guided Learning Module

The AI Guided Learning module is the guided learning section of the application. In this module, students select an anatomy unit and follow the content in a structured order. The system presents anatomical structures one by one so that students do not need to explore the whole model without guidance.

During the learning flow, the currently explained anatomical structure is visually emphasized. Other structures can be made transparent or less dominant to help students focus on the active topic. Information panels are used to show the explanation of the selected structure, and the avatar can support the process by presenting the content through voice.

The module also includes review support. When students need additional clarification, they can return to related structures or request a simpler explanation. This makes the module useful not only for first-time learning but also for repeating difficult topics.

4.2 Free Explore Module

The Free Explore module allows students to inspect anatomical models without following a fixed lesson order. This module is designed for independent exploration and review. Students can focus on the anatomical structures they want to examine and observe them from different angles.

The module supports direct interaction with selected 3D structures. When students point to or interact with anatomical parts, the system can show the name or related information of the selected structure. This helps students connect anatomical terminology with the visual model.

Free Explore complements the AI Guided Learning module by giving students more control over the learning process. After receiving guided explanations, students can use this module to revisit the models and strengthen their spatial understanding.

4.3 AI Chat Module

The AI Chat module allows students to ask anatomy-related questions inside the application. It was implemented as a support module for situations where students need additional explanation beyond the prepared lesson content.

Students can submit questions through text or voice. The generated answer is shown on the interface, and it can also be read aloud depending on the selected voice settings. This makes the module useful in VR, because students can continue looking at the anatomical model while listening to the explanation.

The module is intended to support learning, not to replace teachers or official course materials. Its role is to provide quick clarification, help students review topics, and make the learning experience more interactive.

4.4 Quiz Module

The Quiz module was implemented to help students evaluate their understanding after the learning and exploration stages. The questions are related to the selected anatomy topics included in the current project scope, especially the musculoskeletal and circulatory system contents.

After completing a quiz, students can see their overall performance and review how many mistakes they made in each anatomical topic. The result screen also allows students to understand their topic-based performance through scores or percentage-based feedback. This helps students identify which subjects they understood well and which subjects require further review.

This feature is important for the educational purpose of the application because it connects assessment with learning awareness. Instead of only showing a final score, the system gives students a clearer view of their weak areas.

As a future improvement, the quiz results can be directly connected to the AI Guided Learning module. If a student's success percentage in a topic is below a defined threshold, a review button can be displayed next to that topic. By selecting this button, the student can be redirected to the related AI Guided Learning content and repeat the topic immediately. This would create a stronger connection between assessment and personalized review.

5. Experimental Design

The project includes an experimental design to examine whether the interaction style of the avatar affects the learning experience in a VR anatomy environment. Since the avatar guides students and presents explanations, its communication style may influence student engagement, comfort, motivation, and perceived learning support.

Two experimental conditions were designed for this purpose. Both conditions use the same VR Anatomy application, modules, anatomical content, and core system functions. The difference between them is limited to the level of personalization, the language style, and the avatar presentation. This design makes it possible to focus on the effect of interaction style without changing the main learning content.

At the current stage of the project, only the first experimental condition has been completed with high school students. The second condition is planned as a follow-up study. Therefore, this report presents the completed first condition and the planned second condition, but it does not claim a completed comparison between them.

5.1 Experiment 1: Neutral Interaction Condition

Experiment 1 was implemented as the neutral version of the learning experience. In this condition, the system does not collect or use the student's name. The avatar presents the content with formal instructional language and has a formal visual style.

This condition represents the baseline version of the application. It was used to observe how students interact with the core VR Anatomy system without name-based personalization or a friendlier communication style.

Experiment 1 has been completed with high school students. The feedback and observations collected from this condition are used as the preliminary evaluation of the current project stage.

5.2 Experiment 2: Personalized Interaction Condition

Experiment 2 is planned as the personalized version of the learning experience. In this condition, the system will collect the student's name and use it during the interaction. The avatar will use friendlier and more conversational language, and its visual style will be more casual compared to Experiment 1.

The purpose of this condition is to examine whether personalization and a warmer communication style can make the learning experience feel more engaging and supportive. Since the content and modules remain the same, the comparison will focus on the effect of interaction style.

Experiment 2 has not been conducted yet. After it is completed, its results can be compared with Experiment 1 in terms of engagement, usability, comfort, motivation, and perceived learning support.

6. Evaluation

The evaluation stage of the project focused on observing how high school students interacted with the VR Anatomy application and whether the system supported their understanding of selected anatomy topics. Since only the first experimental condition has been completed so far, the current evaluation is considered a preliminary classroom evaluation based on Experiment 1.

Experiment 1 was conducted with the neutral interaction condition. In this version, the system did not collect student names, used formal instructional language, and presented a formal avatar. The purpose of this evaluation was to examine the usability of the application, the clarity of the learning flow, and the students' initial learning performance after using the VR-based anatomy environment.

The evaluation included both quantitative and observational components. Quantitatively, students completed a short pre-test before using the application and a short post-test after the VR experience. Observationally, students' interaction with the VR environment, their ability to follow the learning flow, their use of the modules, and their general reactions to the application were considered.

The findings from this stage are used to understand the current strengths and limitations of the implemented system and to guide future improvements. Since Experiment 2 has not been conducted yet, this evaluation does not present a comparison between the neutral and personalized interaction conditions.

6.1 High School Testing Process

The first evaluation was carried out with 22 high school students, who represent the target user group of the project. Since VR Anatomy was designed to support introductory anatomy and physiology education, testing the system with high school students was important for understanding its suitability for the intended learning context.

The testing process followed three main steps. First, students completed a five-question pre-test related to the selected anatomy topics. Second, they used the VR Anatomy application under the neutral interaction condition of Experiment 1. During this stage, students experienced the implemented modules, including AI Guided Learning, free exploration, AI Chat, and quiz-based assessment. Third, students completed a five-question post-test at a similar difficulty level.

Before using the application, students were introduced to the VR environment and the basic interaction controls. This step was necessary because some students had limited or no prior experience with VR devices. During the test, students' interactions were observed to identify whether the interface, navigation, model interaction, and learning flow were understandable for the target user group.

The pre-test and post-test design allowed the project team to collect an initial quantitative measure of learning performance. At the same time, observation during the session helped identify usability-related issues such as the need for guidance, comfort in VR, visibility of models, and clarity of interface elements.

6.2 Preliminary Results from Experiment 1

The recorded pre-test and post-test scores indicate a positive change in student performance after using the VR Anatomy application. The average score increased from 2.14 out of 5 in the pre-test to 2.91 out of 5 in the post-test. This corresponds to an average improvement of 0.77 points. In percentage terms, the average score increased from approximately 42.8% to 58.2%, showing an improvement of about 15.4 percentage points.

Individual score changes also showed a generally positive trend. Most students either improved their scores or maintained the same score after using the application. Specifically, 13 students improved their scores, 6 students received the same score, and 3 students received lower scores in the post-test. In addition, the highest score increased from 4 in the pre-test to 5 in the post-test.

These results suggest that the VR Anatomy application may have supported students' short-term understanding of the selected anatomy topics. The increase in the average score indicates that students were able to answer more questions correctly after interacting with the VR learning environment. However, these results should be interpreted as preliminary findings rather than final evidence of learning effectiveness.

Observation during the session also supported the quantitative results. Students were generally interested in the VR-based learning experience, and the use of 3D anatomical models helped make the content more concrete. The guided learning flow, visual highlighting, and avatar-supported explanations helped students focus on the relevant anatomical structures.

6.3 Limitations of the Current Evaluation

Although the preliminary results from Experiment 1 are positive, the current evaluation has several limitations. The most important limitation is that only Experiment 1 has been completed. Since Experiment 2 has not been conducted yet, the project cannot currently provide a direct comparison between the neutral interaction condition and the personalized interaction condition.

Another limitation is the size and scope of the evaluation. The test was conducted with 22 students, which provides useful early feedback but is not sufficient for a broad statistical generalization. A larger participant group would be needed to make stronger claims about the effectiveness of the system.

The post-test was applied shortly after students used the application. Therefore, the results mainly reflect short-term understanding. Long-term retention was not measured in the current evaluation. Future studies should include delayed post-tests to examine whether students remember the content after a longer period of time.

Technical factors may also have affected the evaluation. Since the application includes VR interaction, AI-supported responses, speech processing, and backend communication, the user experience can be influenced by hardware performance, internet connection quality, and backend response time.

For these reasons, the current evaluation should be considered a preliminary classroom evaluation. It provides encouraging early results and useful feedback for future development, but further testing is needed with Experiment 2, larger participant groups, and more systematic data collection.

7. Conclusion

This project presented VR Anatomy, a virtual reality-based educational application designed to support anatomy learning through interactive 3D visualization, avatar-based guidance, artificial

intelligence-supported explanations, and quiz-based evaluation. The main goal of the project was to make introductory anatomy education more visual, engaging, and student-centered by allowing learners to explore anatomical structures in an immersive environment.

Within the current scope, the application focused on selected topics from the musculoskeletal system and the circulatory system. The implemented system includes four main modules: AI Guided Learning, Free Explore, AI Chat, and Quiz. These modules support different stages of the learning process, including guided instruction, independent exploration, question-based clarification, and self-assessment.

The technical implementation combined a Unity-based VR application with a Python-based AI backend. Unity was used to create the 3D learning environment, VR interaction, user interface, avatar presentation, and module flow. The backend supported retrieval-based question answering, speech-to-text, and text-to-speech features. This architecture allowed the system to combine immersive visualization with intelligent learning support.

The first experimental condition was completed with high school students using the neutral interaction version of the application. The preliminary evaluation included pre-test and post-test measurements as well as observations during the VR experience. The results showed an improvement in average student performance after using the application, suggesting that VR Anatomy may support short-term understanding of selected anatomy topics.

However, the evaluation should be interpreted as preliminary rather than final evidence of learning effectiveness. Only Experiment 1 has been completed, while Experiment 2, which includes personalization, friendlier language, and a more casual avatar, is planned as a follow-up study. Therefore, a direct comparison between the two interaction conditions has not yet been completed.

Overall, the project demonstrates that VR and AI technologies can be combined to create a more interactive and supportive anatomy learning environment. VR Anatomy provides a functional prototype that can be expanded with broader anatomical content, stronger personalization, improved evaluation methods, and more advanced adaptive learning features in future work.

8. Future Work

Although VR Anatomy provides a functional prototype for VR-based anatomy learning, there are several directions for future improvement. These improvements can strengthen both the educational effectiveness and the technical quality of the system.

The first future direction is to complete Experiment 2 and compare it with Experiment 1. Experiment 2 will include name-based personalization, friendlier language, and a more casual avatar. After this condition is tested with students, the results can be compared with the neutral interaction condition in terms of engagement, usability, comfort, motivation, and perceived learning support. This comparison will help determine whether personalization improves the VR anatomy learning experience.

A second improvement is to conduct the evaluation with a larger participant group and a more detailed experimental design. The current evaluation provides preliminary results, but future studies should include more students, longer testing sessions, and more systematic data collection.

In addition, delayed post-tests can be used to measure whether students remember the learned content after a longer period of time.

The quiz module can also be improved by connecting quiz results directly to the learning modules. In the current version, students can see their topic-based performance after completing a quiz. In a future version, if a student's success percentage in a topic is below a defined threshold, a review button can appear next to that topic. By selecting this button, the student can be redirected to the related content in the AI Guided Learning module and repeat the topic immediately. This would create a stronger connection between assessment and personalized review.

Another future direction is to expand the anatomical content of the application. The current version focuses on selected topics from the musculoskeletal and circulatory systems. Future versions can include additional body systems such as the respiratory system, nervous system, digestive system, urinary system, and sensory organs. Expanding the content would make the application more complete and useful for broader anatomy education.

Technical improvements are also necessary for future versions. These include optimizing 3D models for better VR performance, reducing backend response time, improving speech-to-text and text-to-speech stability, and making the application more reliable on different devices. Avatar animation and lip-sync quality can also be improved to make the learning experience more natural.

Finally, future versions can include teacher-oriented features. For example, a teacher dashboard can be developed to show student progress, quiz results, frequently missed topics, and general usage statistics. This would allow teachers to use VR Anatomy not only as a student learning tool, but also as a classroom monitoring and support system.